“Cyber Safety Awareness and Security Steps for Safer Online Use”

A Dissertation By
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DECLERETION

I hereby declare that the thesis entitled "**Cyber Safety Awareness and Security Steps for Safer Online Use**"submitted to the Department of Computer Science and Engineering (CSE), Pundra University of Science and Technology, is a record of original work carried out by me under the supervision of Md. Polash Tai. This research has not been submitted previously, either in part or in full, to any other institution or university for the award of any degree, diploma, or other qualification.

All information, data, references, and sources used in this thesis have been duly acknowledged and cited in accordance with academic and ethical standards.I take full responsibility for the authenticity and accuracy of the contents presented in this document.

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ABSTRUCT

In the contemporary digital era, the extensive reliance on information technology has woven itself into virtually every aspect of daily life. However, despite the ubiquity of online platforms, a significant portion of users lack adequate awareness regarding cyber safety, exposing themselves to considerable risks such as data breaches, identity theft, and cyber fraud. This study systematically examines user awareness, behavioral patterns, and the security measures adopted to mitigate cyber threats.The primary objective is to identify the gaps in knowledge and practice among general users and to propose effective, user-friendly security steps that can be easily implemented to enhance online safety. Employing quantitative survey methods alongside behavioral analysis, the research highlights critical vulnerabilities and presents actionable recommendations tailored to diverse user groups including students, professionals, and technology-dependent individuals. Ultimately, this study aims to contribute to the development of safer digital environments by empowering users with the necessary knowledge and tools to foster responsible and secure online behavior.

TABLE OF CONTENT

Chapter 01 : Introduction

1.1 Background of the Study

In today's digitally interconnected world, individuals are increasingly reliant on internet-based platforms for communication, transactions, and data management. While this digital evolution offers numerous benefits, it also brings serious challenges in terms of cyber security. Although many technological tools are available for digital protection—such as antivirus software, firewalls, and secure authentication systems—human behavior remains the weakest link in the security chain. Many users still lack awareness of basic online safety practices, making them susceptible to a range of cyber threats, including identity theft, malware, phishing attacks, and data breaches.

1.2 Problem Statement

Despite advancements in digital security technologies, a significant portion of users fails to adopt proper cyber hygiene practices. This behavioral gap stems largely from a lack of awareness, negligence, or misconceptions about the relevance and ease of implementing security measures. As a result, individuals unknowingly engage in risky activities such as using weak passwords, sharing personal data carelessly, and ignoring system updates, thereby increasing their vulnerability. This research seeks to explore why users do not adopt safe digital behaviors, even when tools and knowledge are available.

1.3 Research Objectives

The main objective of this study is to investigate user behavior and awareness related to cyber safety. Specific objectives include:

- To assess the level of awareness about cyber threats and safety measures.
- To analyze common risky behaviors in digital environments.
- To identify gaps between awareness and actual practice.
- To propose practical strategies for enhancing individual cyber safety.

1.4 Research Questions

This research is guided by the following questions:

- What is the current level of awareness among users regarding cyber hygiene?
- What behavioral patterns increase vulnerability to online threats?
- Why do users neglect available security tools despite understanding the risks?
- What measures can be recommended to improve cyber-safe behavior?

1.5 Scope of the Study

The study focuses primarily on individual internet users in both urban and rural contexts, especially those who are digitally active but lack advanced technical knowledge. It examines their awareness levels, daily online habits, and the effectiveness of currently used protective measures. The research does not delve into enterprise-level cybersecurity systems but remains focused on user-end vulnerabilities and practices.

1.6 Significance of the Study

The importance of this study lies in its user-centric approach to digital safety. While many studies focus on technical solutions, this research highlights the role of individual behavior in enhancing or compromising cyber security. The findings will not only contribute to academic discussions but also offer actionable recommendations for awareness campaigns, educational programs, and policy interventions aimed at promoting safer online habits among general users.

Chapter 02 : L iterature Review

2.1 Introduction

The last few decades have witnessed unprecedented growth in information technology, deeply embedding digital platforms into virtually every aspect of human life. Activities such as online communication, financial transactions, education, and employment have become increasingly internet-dependent. However, this rapid digitization has also amplified the complexity and frequency of cyber threats worldwide. Although technological advancements and policy frameworks are indispensable in countering cyberattacks, human factors—particularly user awareness and behavioral practices—remain the most crucial yet vulnerable components of cybersecurity. This study offers an integrative examination of four core dimensions of cyber safety: the extent and depth of user awareness as a foundation for secure online conduct; the efficacy of current security technologies and protocols; behavioral tendencies of users that influence their exposure to risks; and the varied nature and consequences of online threats. By moving beyond purely technical solutions, this research emphasizes the pivotal role of human-centric cyber hygiene. Ultimately, it aims to develop actionable strategies that empower diverse user groups—ranging from laypersons to professionals—to cultivate safer and more resilient digital environments.

2.2 User Awareness and Behavioral Dynamics

In the fast-evolving digital milieu, the nexus between user awareness and behavior is fundamental to personal cybersecurity. International research reveals stark disparities in awareness and security-related conduct, which critically determine susceptibility to cyber risks [1][2][3]. Many users possess a rudimentary grasp of cybersecurity principles but often fail to apply best practices consistently. Common shortcomings include poor password hygiene, reliance on weak credentials, and vulnerability to phishing attempts via suspicious emails or links, all of which substantially elevate risk [4][12][13]. Conversely, younger and more tech-savvy individuals, despite higher awareness, frequently exhibit risky behaviors such as excessive disclosure of personal data and noncompliance with cybersecurity protocols [5][6]. Additionally, professionals across sectors, while cognizant of security imperatives, may neglect strict adherence due to time constraints or limited technical expertise, thereby increasing both personal and organizational vulnerabilities [7]. Evidence strongly suggests that enhancing user awareness through tailored education and targeted training can significantly improve adherence to cybersecurity best practices [8][9]. Therefore, developing specialized awareness programs customized to heterogeneous user profiles is vital for mitigating cyber threats effectively.

2.3 Security Mechanisms and Best Practices

The adoption of appropriate security mechanisms is indispensable for fostering safe online behavior at the individual level. Empirical studies indicate that a majority of cybersecurity breaches originate not from sophisticated attacks but from user negligence or lack of awareness about fundamental protective measures [5]. Security interventions can broadly be classified into technological defenses and behavioral adaptations. Technological safeguards include deploying robust, unique passwords, implementing two-factor authentication (2FA), employing end-to-end encryption, utilizing secure browsers, and maintaining up-to-date software systems [6]. Notably, enabling 2FA can reduce the risk of account compromise by as much as 70% [7]. Protective software tools are effective against phishing, malware, spam, and social engineering threats [8]. Despite this, a significant portion of users engage in risky behaviors such as transmitting sensitive information over unsecured public Wi-Fi networks, thereby exposing themselves to attacks [9]. Employing Virtual Private Networks (VPNs) and updated antivirus programs is recommended to mitigate these risks.

The literature also underscores the importance of integrating security features at the design phase—a principle known as privacy by design—which shifts the security burden from users to the system architecture itself [10]. This proactive approach promotes inherently secure digital environments rather than relying solely on end-user vigilance.

Furthermore, cybersecurity education is paramount. Many users underestimate the risks associated with installing unverified mobile applications or sharing sensitive information online. Interactive tutorials, digital safety courses, and virtual workshops have proven effective in enhancing user competence and confidence [11]. However, some studies highlight a usability-security dilemma, especially among elderly populations and individuals with limited digital literacy, where overly complex security protocols discourage adoption and inadvertently increase risk exposure [12]. This necessitates the development of user-centric security solutions balancing robustness and accessibility.

In sum, fortifying individual cybersecurity resilience demands a dual approach—deploying technological safeguards and cultivating an informed, cautious digital culture—to establish a safer online ecosystem for all users.

2.4 User Behavior: Its Influence on Cybersecurity

User behavior critically shapes the effectiveness of cybersecurity measures, as advanced technical defenses may be undermined by negligent or uninformed practices. Research consistently documents prevalent risky behaviors including weak password management, indiscriminate clicking on dubious links, and oversharing of personal information, all of which degrade users' security posture [11][12].

Nguyen and Tran (2019) highlight that behavioral challenges underlie widespread non-compliance with security protocols, such as reluctance to enable 2FA or regularly update passwords, thus elevating vulnerability to cyber intrusions [11]. Patel and Desai (2021) demonstrate that convenience often takes precedence over caution, with users sharing passwords or connecting to unsecured public Wi-Fi, increasing threat exposure [12]. Ahmed and Rahman (2020) reveal demographic differences in behavior, where tech-savvy youth display immature security habits despite technical competence, whereas older or less literate users suffer from limited awareness leading to inadvertent security breaches [13]. Garcia and Silva (2020) find that employees in corporate settings, though aware of security importance, often neglect compliance due to workload and perceived inconvenience, jeopardizing organizational data integrity [14]. Zhao and Wang (2021) examine social media's role in privacy risks, noting that oversharing personal data substantially amplifies user vulnerability within digital ecosystems [15].

2.5 Summary of Key Findings from Literature

The literature review reveals a multi-dimensional understanding of cyber hygiene shaped by user awareness, behavior, and institutional security measures. The concept of cyber hygiene has evolved beyond traditional notions of antivirus use and has become closely linked with proactive digital behavior, including secure password practices, regular system updates, and cautious interaction with online content. The reviewed studies consistently indicate that user awareness is fragmented—particularly in rural or less-educated populations—where fundamental digital literacy and online safety practices remain low.

Behavioral analysis shows that despite increasing digital adoption, user actions often contradict basic security principles, driven by a lack of awareness or convenience-seeking habits. The literature also highlights critical vulnerabilities such as social media oversharing, weak authentication, and insufficient use of cybersecurity tools. Current security infrastructures are generally reactive, with most users relying on basic protection (like free antivirus software) without understanding the depth of emerging cyber threats like phishing, ransomware, or zero-day vulnerabilities.

Furthermore, socio-demographic factors such as age, education, gender, and digital exposure significantly influence awareness and cyber behavior. Women, the elderly, and individuals outside formal education or urban environments are disproportionately exposed to digital risks due to limited digital training or awareness programs. Collectively, the literature suggests an urgent need for customized awareness campaigns, inclusive cyber hygiene education, and scalable technological interventions to bridge the behavioral and knowledge gaps. This synthesized understanding establishes the groundwork for identifying the research gaps and developing strategies aimed at fostering safer digital practices across diverse user groups.

Chapter 03 : Research Gap and Limitations

3.1 Identified Gaps in Prior Studies

A review of the existing literature reveals that many prior studies have largely concentrated on narrow or specific user groups, such as students, corporate employees, or users in urbanized regions [5][6]. This focus has inadvertently created a critical knowledge gap by omitting a more holistic portrayal of users with varied demographic and socio-economic backgrounds. As a result, the complex interrelationships between user awareness, technological preparedness, and behavioral adaptability across diverse communities remain significantly underexplored.

Despite a considerable body of research addressing technical security frameworks, there is an evident deficiency in examining how behavioral, psychological, and cultural dynamics shape cyber safety practices [7]. The depth of understanding regarding how different social groups perceive and react to cybersecurity threats—especially in terms of educational attainment, digital literacy, and professional exposure—remains largely inadequate. Further, studies have yet to fully address how evolving threats such as AI-driven attacks, machine learning–enabled phishing, and socially engineered malware campaigns impact diverse populations, particularly those lacking advanced digital proficiency [22].

Another overlooked dimension involves the role of psychological and cultural constructs in influencing cybersecurity behavior. For example, user habits, cultural norms, and even fatalistic attitudes toward online risks may undermine the adoption of secure practices despite awareness. Without incorporating such variables, existing frameworks fail to offer actionable or inclusive security strategies. Therefore, these omissions point to the urgent need for a comprehensive investigation that bridges technical safeguards with human-centric cyber behavior across a wider spectrum of user identities.

3.2 Target Audience and Data Limitations

An additional limitation arises from the underrepresentation of vulnerable or less digitally literate populations in previous surveys and studies. These include rural communities, women engaged in informal sectors, elderly citizens, and individuals with low educational attainment—groups that often remain outside the digital inclusion radar. Their exclusion leads to an incomplete understanding of the nation's cyber hygiene landscape.

Moreover, earlier data collection methods have relied heavily on online platforms, academic institutions, or tech-savvy users, thereby inadvertently excluding users with minimal internet exposure. As a result, a significant segment of the population—particularly those most at risk of exploitation—has remained underrepresented in cybersecurity research. Without data from these groups, the reliability and inclusiveness of cyber safety policy recommendations become questionable. This limitation is particularly critical in the context of developing regions, where infrastructural and awareness gaps compound cybersecurity challenges.

3.3 Need for New Field-Level Data

To address these gaps, it is imperative to design a research strategy that emphasizes field-level data collection from heterogeneous user groups. The proposed study aims to integrate data from both digitally active users and those on the margins of digital engagement. Field-level interactions, interviews, and context-aware surveys will be utilized to obtain ground-level insights into users' behaviors, awareness levels, and responses to cyber threats.

Such an inclusive data-driven approach will enable the formulation of a more accurate and adaptive cybersecurity framework—one that is not only technologically robust but also socially and behaviorally informed. By contextualizing risk perception, behavioral readiness, and awareness across diverse communities, the study seeks to redefine cyber hygiene strategies in a manner that aligns with both global threat landscapes and local socio-cultural realities [9][10].

Chapter 04 : Research Methodology

4.1 Research Design and Rationale

This study employs a mixed-methods research design to explore cybersecurity awareness, user behavior, and risk management practices among digitally active and vulnerable populations in Bangladesh. The rationale behind this approach lies in the diverse digital literacy levels across the country's rural, semi-urban, and urban communities. Given the heterogeneity in technological exposure, a combination of quantitative and qualitative methods ensures a holistic understanding of end-user cybersecurity practices. This research design facilitates triangulation of data and enables contextual interpretation, which is particularly essential in developing actionable interventions for cyber hygiene enhancement.

4.2 Target Audience Selection

To ensure demographic diversity and relevance, the study strategically selects three primary target groups based on their digital engagement and socio-cultural positioning:

- Rural and Low-Literacy Users: Individuals with minimal formal education and limited access to digital infrastructure, often reliant on basic mobile phone use. This group reflects a high-risk segment due to their limited cybersecurity awareness and lack of access to protective tools or education.
- Secondary School Students (Ages 13–18): Young users from both rural and urban schools who regularly interact with digital devices and services. This group is in the developmental phase of cyber awareness, making them critical for preventive education strategies.
- Urban Teachers and Social Workers: Educated professionals engaged in digital communication and education, often acting as intermediaries for disseminating knowledge and best practices. Their insights can provide both reflective and prescriptive value to the study.

Each group will include approximately 50 participants, bringing the total sample size to 150, thereby ensuring equitable representation and data richness.

4.3 Sampling Strategy

The research adopts a purposive sampling strategy, allowing the deliberate selection of participants who meet specific inclusion criteria related to digital exposure, socio-educational background, and regional location. This strategy is particularly suitable for exploratory studies where representativeness and contextual depth outweigh statistical generalizability. Equal representation across the three groups enhances the study's capacity to identify comparative trends and segment-specific challenges.

4.4 Data Collection Tools and Techniques

To address the varying levels of literacy and technological familiarity, two complementary data collection tools are deployed:

A. Structured Questionnaire

A closed-format questionnaire designed for literate participants (students, teachers, and social workers). It includes:

- Multiple-choice questions
- 5-point Likert-scale items
- Awareness of cyber threats
- Protective behaviors (e.g., password practices, antivirus use)
- Daily online activities
- Experience with cybersecurity training
- The questionnaire is pilot-tested and validated to ensure clarity, linguistic suitability, and cultural appropriateness.

B. Semi-Structured Interviews

- Designed primarily for rural and low-literacy respondents, the interviews:
- Capture in-depth narratives
- Allow flexibility in responses
- Explore contextual factors influencing cybersecurity behavior

An interview guide ensures consistency across interviews while allowing for the exploration of emergent themes, emotional responses, and personal experiences.

4.5 Questionnaire and Question Areas

Question domains were developed based on a thorough literature review and adapted from national cybersecurity policy recommendations and international cyber hygiene frameworks. Key areas include:

- General cybersecurity knowledge
- Risk perception and threat awareness
- Behavioral patterns and device usage
- Experience with phishing or scams
- Use of public Wi-Fi, USBs, and online platforms
- Access to training or support

This ensures content validity and relevance to both the national context and global best practices.

4.6 Duration and Execution Plan

Activity Duration

- Questionnaire & Interview Guide Development 1 week
- Data Collection 4 weeks
- Data Cleaning & Statistical Analysis 2 weeks
- Report Writing & Final Review 1–2 weeks

The fieldwork will be conducted in both urban and rural regions, with coordination from local institutions to ensure access and participant engagement.

4.7 Ethical Considerations

The study strictly adheres to ethical research principles. Key protocols include:

- Informed Consent: All participants will be briefed on the study's objectives, procedures, and their rights, with written consent obtained.
- Confidentiality: Data will be anonymized and stored securely to protect participant identity.
- Ethical Clearance: Approval has been obtained from the institutional ethics review board in compliance with national and international research standards.
- Voluntary Participation: Respondents retain the right to withdraw at any time without consequence.

4.8 Data Analysis Procedures

A. **Quantitative Data:**

Data from structured questionnaires will be analyzed using SPSS and Microsoft Excel. Analytical procedures include:

- Descriptive statistics (mean, frequency, percentage)
- Cross-tabulations to identify group-based patterns
- Correlation and regression analysis for inferential insights

B. **Qualitative Data**:

Interview transcripts will undergo thematic analysis using open and axial coding. This will allow the identification of:

- Recurrent themes
- Behavior drivers
- Cultural influences
- Barriers to cybersecurity adoption

4.9 Limitations

The research acknowledges several inherent limitations:

- Digital Literacy Variance: Differential understanding among participants may affect the reliability of self-reported data.
- Sampling Constraints: Purposive sampling may limit generalizability.
- Geographical Dispersion: Travel and logistical constraints could restrict data collection in certain remote areas.
- Response Bias: Participants may provide socially desirable answers, especially regarding security behaviors.

4.10 Expected Contributions

This study is anticipated to yield the following academic and practical contributions:

- Empirical Insights: Generate original field-level data on cybersecurity awareness in Bangladesh.
- Policy Implications: Inform government and non-profit initiatives on targeted cyber hygiene interventions.
- Educational Design: Support the creation of culturally appropriate training programs for at-risk user groups.
- Academic Foundation: Provide a structured baseline for future cybersecurity behavior studies in South Asia.

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